

Quartile Deviation

$$Q.D = \frac{Q_3 - Q_1}{2}$$

$$D_5 = L + \frac{h}{f} \left(5 \frac{\sum f}{10} - Cf \right)$$

$$Q_3 = L + \frac{h}{f} \left(3 \frac{\sum f}{4} - Cf \right)$$

$$P_{90} = L + \frac{h}{f} \left(90 \frac{\sum f}{100} - Cf \right)$$

$D_5 \rightarrow$ Ungrouped $\rightarrow$ odd

$$\frac{5(n+1)}{10}$$

FINAL TERM

6/1/25

Regression Equation
(b hat)

$$y = b_0 + b_1 x$$

$$b_1 = \frac{\sum xy - (\sum x)(\sum y)/n}{\sum x^2 - (\sum x)^2/n}$$

$$b_0 = \frac{1}{n} (\sum y - b_1 \sum x)$$

} $n \rightarrow$ # of column

10

Graph $\rightarrow$ Min per circle (o) Max per circle (.)

$y = 195 - 20x$
 $\rightarrow$ negative, Age increase, Price Decrease

Prediction mean x mean value put

One-Line Regression

`model = LinearRegression().fit(X,y)` Assign 3

Sum of Square Regression

$$SST = \frac{\sum y^2 - (\sum y)^2}{n} \quad (\text{Total Sum of Square})$$

Covariation

$$r^2 = \frac{SSR}{SST}$$

1 $\rightarrow$ Perfect

0.8 $\rightarrow$ Positive Strong

`r_squared = LinearRegression().fit(x,y).score(X,y)`

13/1/25

Sample $\rightarrow$ With Replacement
 Without Replacement

Sample Mean = Population Mean

10

Graph $\rightarrow$ Min per circle ($\circ$) , Max per circle ($\bullet$)

Date: _____

$$y = 195 - 20x$$

$\rightarrow$ negative

Age

Price increase, Price Decrease

Prediction mein x mean value put

One-Line Regression

`model = LinearRegression().fit(X,y)` Assign 3

Sum of Square Regression

$$SST = \frac{\sum y^2 - (\sum y)^2}{n} \quad (\text{Total Sum of Square})$$

Covariation

$$r^2 = \frac{SSR}{SST}$$

1 $\rightarrow$ Perfect

0.8 $\rightarrow$ Positive Strong

`r_squared = LinearRegression().fit(x,y).score(X,y)`

13/1/25

Sample $\rightarrow$ With Replacement
Without Replacement

Sample Mean = Population Mean

Without Replacement = ${}^N C_A$

Date: _____

○ 4 8 12 → Population → 4
Sample Size.

✓ Sample Mean ⇒

$$\mu = \sum \bar{x} \cdot P(\bar{x})$$
$$\sigma_{\bar{x}}^2 = \sum \bar{x}^2 \cdot P(\bar{x})$$

Sample Mean
Population Mean
Sample S.D
Population S.D
Standard Error

2 SAME

$N \rightarrow$ Population

$n \rightarrow$ Sample

$$\text{Standard Error} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

With Replacement

$(N)^n$
Ans → 16

(Pop) Sample Size
 $P(\bar{x}) = \frac{1}{16}$

$$\text{Standard Error} \rightarrow \frac{\sigma}{\sqrt{n}}$$

Date: _____

RANDOM VARIABLE

Associate a real number assign

Probability Mass Function

$$P_{xi} \geq 0, \sum P(P_{xi}) = 1$$

Discrete mean	Summation
Continuous mean	Integration

16/1/25

Binomial Distribution

~~Multi Binomial Distribution~~

Multinomial Distribution

Poisson Distribution

Negative Binomial Distribution

Geometric Distribution

Sampling Distribution W / Rep
Without Replace

Random Variable

Discrete $\rightarrow$

Probability Mass Function

Probability Density Functions

Random Variable

Conditional Probability

Date: _____

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

given that $\rightarrow$ Conditional
 $\hookrightarrow$ Denominator

General Multiplication Rule

$$P(A \cap B) = P(A) \cdot P(B|A)$$

- 1) Conditional Probability $\rightarrow$ Walpole $\rightarrow$ 2.6
- 2) Concept of Dis/Cont Random Var, Distribution & Density Func
- 3) Joint Density & Distribution Function
- 4) Discrete Probab
- 5) Poisson Distribution

HYPOTHESIS

- 1) Null, Alternate Hypothesis
 - 2) Significance Level
 - 3) Test Statistics
 - 4) Critical Region
 - 5) Reject Rule
 - 6) Conclusion
- more than, less, equal

Extra Class

2 hour Class

Date: _____

① Null Alternative $\mu = \mu_0$ NE More than less than

② Significance $\rightarrow$

test mein $z \rightarrow$ $t \rightarrow$ Nhi hitna

Discrete

Continuous $\rightarrow$ N.D $\rightarrow \frac{x - \text{mean}}{\text{Stand Dev}}$

20/11/25

11, 12X

CLO 3 $\rightarrow$ Sampling, Hypothesis, Goodness of fit test

Cumulative Distribution Function

CLO 1 $\rightarrow$ 6 Marks

CLO 2 $\rightarrow$ ~~Basic Con~~ Conditional Prob Density
Discrete / Continuous, Random Variable, Dist^y
X 11 - 12X

Disc

Q - 14 Mobee CIS

X 11 - 12X

$$\begin{array}{r} -0.56 \\ -0.5 \end{array} \quad 0.06$$



Date: _____

CLO-3 →

20 Marks

Hypothesis, Sampling distri, Estimation Theory

Goodness of fit test
fit the expected distributions

Confirm?

Chi Squared

6 steps